



This problem will explore dark matter freeze-out in the early universe. We will derive approximate analytic expressions for the relic abundance of a heavy particle that was once in thermal equilibrium with the primordial plasma and froze out while non-relativistic. This is the classic “WIMP” (weakly interacting massive particle) scenario for dark matter production.

Consider a heavy dark matter particle χ (mass m_χ) and its antiparticle $\bar{\chi}$ in the early Universe. They can annihilate into a pair of light Standard Model (SM) particles $\psi_{\text{SM}}, \bar{\psi}_{\text{SM}}$,

$$\chi + \bar{\chi} \longleftrightarrow \psi_{\text{SM}} + \bar{\psi}_{\text{SM}}. \quad (0.1)$$

A few points to note:

- The light species $\psi_{\text{SM}}, \bar{\psi}_{\text{SM}}$ remain in full thermal and chemical equilibrium with the radiation bath at all times.
- There is no particle-antiparticle asymmetry in the χ sector: $n_\chi = n_{\bar{\chi}}$.
- The universe is radiation dominated during freeze-out.

Throughout, take the effective relativistic degrees of freedom in energy and entropy to be equal and approximately constant: $g_* \simeq g_{*S} \simeq 10$.

1 Equilibrium abundance of a non-relativistic species (2 points)

- (a) Starting from the definition of the number density from the distribution function $f(p)$, show that the number density of a non-relativistic species χ in thermal and chemical equilibrium is approximately

$$n_\chi^{\text{eq}}(T) \simeq g \left(\frac{m_\chi T}{2\pi} \right)^{3/2} e^{-m_\chi/T}, \quad (1.1)$$

where g is the number of internal degrees of freedom of χ (e.g. $g = 4$ for a Dirac fermion) and m_χ is its mass. Does this result depend on whether χ is a boson or a fermion?

- (b) With the entropy density of the plasma,

$$s(T) = \frac{2\pi^2}{45} g_{*S} T^3, \quad (1.2)$$

we define the comoving equilibrium abundance

$$N_\chi^{\text{eq}} \equiv \frac{n_\chi^{\text{eq}}}{s}. \quad (1.3)$$

Derive an explicit expression for $N_\chi^{\text{eq}}(T)$ in terms of $x \equiv m_\chi/T$, g , and g_{*S} , and show that for large x ,

$$N_\chi^{\text{eq}}(x) \propto x^{3/2} e^{-x}. \quad (1.4)$$

2 Simplifying the Boltzmann equation (4 points)

The Boltzmann equation for the number density of X in an expanding FRW Universe can be written as

$$\frac{dn_\chi}{dt} + 3Hn_\chi = -\langle\sigma v\rangle (n_\chi^2 - (n_\chi^{\text{eq}})^2). \quad (2.1)$$

Recall, n_χ is the comoving number density of χ particles in the universe while n_χ^{eq} is the comoving number density in thermal and chemical equilibrium. It is not true in general that $n_\chi = n_\chi^{\text{eq}}$, but that is precisely what we want: n_χ should depart from n_χ^{eq} as the universe expands and cools, such that eventually the annihilation rate becomes negligible and n_χ “freezes out” to a constant comoving abundance.

- (a) Using the definition $N_\chi \equiv n_\chi/s$ and the fact that in adiabatic expansion $sa^3 = \text{const}$, show that the Boltzmann equation can be written as

$$\frac{dN_\chi}{dx} = -\frac{s}{Hx} \langle\sigma v\rangle (N_\chi^2 - (N_\chi^{\text{eq}})^2). \quad (2.2)$$

where H is the Hubble parameter and $x \equiv m_\chi/T$. (Hint: $a \propto T^{-1}$ during radiation domination, so $\frac{dx}{dt} = Hx$.)

- (c) For a radiation-dominated Universe, the Hubble parameter is

$$H(T) = \sqrt{\frac{4\pi^3}{45}} \frac{\sqrt{g_*} T^2}{M_{\text{Pl}}}, \quad (2.3)$$

with reduced Planck mass $M_{\text{Pl}} \simeq 2.4 \times 10^{18} \text{ GeV}$. Evaluate $s/(Hx)$ at general temperature and then at $T = m_\chi$ (i.e. $x = 1$) to define a constant

$$\lambda \equiv \frac{s(m_\chi) \langle\sigma v\rangle}{H(m_\chi)}. \quad (2.4)$$

Show that Eq. (2.2) can be written in the “Riccati” form

$$\frac{dN_\chi}{dx} = -\frac{\lambda}{x^2} (N_\chi^2 - (N_\chi^{\text{eq}})^2). \quad (2.5)$$

Give λ explicitly in terms of g_* , g_{*S} , m_χ , M_{Pl} , and $\langle\sigma v\rangle$.

3 Approximate analytic solution (2 points)

The Riccati equation (2.5) cannot be solved in closed form with the exact equilibrium abundance, but it admits a standard approximate solution using the *freeze-out approximation*.

At early times (small x), interactions are rapid and $N_\chi \approx N_\chi^{\text{eq}}$. In this regime the interaction rate per particle $\Gamma \equiv n_\chi^{\text{eq}} \langle\sigma v\rangle$ satisfies $\Gamma \gg H$.

- (a) Define the freeze-out point x_f by the condition

$$\Gamma(x_f) \approx H(x_f) \quad \Rightarrow \quad n_\chi^{\text{eq}}(x_f) \langle \sigma v \rangle \approx H(x_f). \quad (3.1)$$

Using your expressions from Problems 1 and 2, derive an implicit equation determining x_f :

$$x_f \simeq \ln \left[c(c+2) \frac{g}{\sqrt{g_*}} \frac{m_\chi M_{\text{Pl}} \langle \sigma v \rangle}{x_f^{1/2}} \right], \quad (3.2)$$

where c is an $\mathcal{O}(1)$ numerical constant introduced to refine the criterion (you may take $c \approx 1-1.5$). *Hint:* Start from Eq. (3.1), plug in the explicit forms of n_χ^{eq} and H , and isolate logarithms of x_f .

- (c) After freeze-out, N_χ^{eq} becomes exponentially small, and the Boltzmann equation (2.5) simplifies to

$$\frac{dN_\chi}{dx} \approx -\frac{\lambda}{x^2} N_\chi^2, \quad (x \gtrsim x_f). \quad (3.3)$$

Solve this differential equation analytically for $N_\chi(x)$ for $x \geq x_f$, imposing the matching condition $N_\chi(x_f) \approx N_\chi^{\text{eq}}(x_f)$, and obtain the asymptotic abundance

$$N_\chi(\infty) \equiv N_\infty \approx \frac{x_f}{\lambda}. \quad (3.4)$$

Why is N_∞ inversely proportional to the annihilation cross section $\langle \sigma v \rangle$?

4 Relic density and the “WIMP miracle” (2 points)

- (a) The present-day dark matter density in units of the critical density is

$$\Omega_\chi h^2 = \frac{\rho_\chi}{\rho_c} h^2 \simeq \frac{m_\chi n_\chi^0}{\rho_c/h^2} = \frac{m_\chi s_0 N_\infty}{\rho_c/h^2}, \quad (4.1)$$

where $s_0 \simeq 2890 \text{ cm}^{-3}$ is today’s entropy density and $\rho_c/h^2 \simeq 1.05 \times 10^{-5} \text{ GeV cm}^{-3}$ (h is the dimensionless Hubble parameter). Note we used $n_\chi^0 = s_0 N_\infty$ since N_χ is comoving and thus constant after freeze-out. Using your result for N_∞ , show that

$$\Omega_\chi h^2 \approx \frac{0.1 \text{ pb}}{\langle \sigma v \rangle} \quad (4.2)$$

up to $\mathcal{O}(1)$ factors and logarithms, where $1 \text{ pb} = 10^{-36} \text{ cm}^2$.
(Hint: Convert $\langle \sigma v \rangle$ from GeV^{-2} to cm^3/s as needed.)

- (b) Take the observed dark matter abundance to be $\Omega_{\text{DM}} h^2 \simeq 0.12$. For a WIMP with mass $m_\chi = 100 \text{ GeV}$, estimate the required annihilation cross section $\langle \sigma v \rangle$. Compare your answer to a “typical” weak-interaction cross section. Explain why this is often referred to as the *WIMP miracle*.

5 Numerical solution (2 bonus points)

This problem is optional and intended as a programming exercise (e.g. in Python, Mathematica, or C++).

- (a) Starting from the Riccati equation

$$\frac{dN_\chi}{dx} = -\frac{\lambda}{x^2} (N_\chi^2 - (N_\chi^{\text{eq}})^2), \quad (5.1)$$

implement a numerical ODE solver to evolve $N_\chi(x)$ from an initial point $x_{\text{init}} \ll 1$ (e.g. $x_{\text{init}} = 1$) to a large final value $x_{\text{final}} \gg x_f$ (e.g. $x_{\text{final}} = 1000$).

- (b) Compare your numerical solution for $N_\chi(x)$ with:

- 1) the equilibrium abundance $N_\chi^{\text{eq}}(x)$, and
- 2) the analytic freeze-out approximation from Problems 3–4.

Plot all three curves on the same graph and identify the freeze-out epoch.

You can also vary $\langle\sigma v\rangle$ over at least two orders of magnitude and study how the final relic abundance changes. Confirm the inverse proportionality $\Omega_\chi \propto 1/\langle\sigma v\rangle$ to good accuracy.